Applied Question: Social Network Analysis and Contact Tracing

You are a Public Health and Preventive Medicine resident working with a local public health unit investigating a prolonged pertussis outbreak among adolescents in a mid-sized urban community. Routine case and contact management has not stopped transmission. During follow-up interviews, confirmed cases report close contacts over the previous 21 days (defined as ≥15 minutes of face-to-face interaction or shared enclosed space).

Contact Tracing Information

Below is a simplified contact-tracing chart based on interviews with the first five confirmed cases.

Person ID	Case Status	Close Contacts Reported
A	Confirmed case	B, C, D
B	Confirmed case	A, C, E
C	Confirmed case	A, B, D, F
D	Confirmed case	A, C, G
E	Confirmed case	B, F
F	Not a case	C, E, H
G	Not a case	D
H	Not a case	F

Assume all contacts are bidirectional, and no other links are known.

Questions

1. Using the contact-tracing chart above, draw a social network diagram representing the relationships between individuals A–H. Clearly label each node and indicate confirmed cases versus non-cases (e.g., by shading or shape).
2. Based on your network diagram, identify one individual with high degree centrality and one individual with high betweenness centrality. Briefly justify your choices using features of the network.
3. From a transmission-control perspective, identify two individuals or groups that should be prioritised for targeted public health interventions (e.g., vaccination catch-up, prophylaxis, exclusion, or enhanced follow-up). Explain your rationale using concepts from social network analysis.
4. After reviewing your analysis, the Medical Officer of Health asks whether this outbreak appears to be driven by a core group, bridging individuals, or household-style clustering. Based on the network, which pattern best fits and why?